

Human Action Detection

2D Classification | 3D Video Models | YOLO Detection + Web App

EfficientNet-B3 @300px: 85.24% test

R3D-18 on KTH: 91.6% validation

Stack: PyTorch + YOLOv8 + Gradio | RTX 5060

1. Introduction and Objectives

ActionScope is an end-to-end human action detection system in PyTorch with three branches: a 2D frame classifier for 15 everyday actions, a true-temporal 3D CNN for 6 video actions, and YOLOv8 person detection fused with the 2D head for per-person labels. A premium Gradio app serves all three.

- Classify 15 routine activities in still images.
- Model motion over time with 3D CNNs on real clips.
- Detect people (YOLO) and label each person's action.
- Ship a polished demo, honest metrics, and deployment artifacts.

2. Datasets (Kaggle)

Dataset	Size	Classes	Split
HAR-15 (images)	~218 MB, 12,600 imgs	15	10,710 / 1,890
KTH (video)	600 clips, 25 subj.	6	train / val (80/20)

HAR-15 classes: calling, clapping, cycling, dancing, drinking, eating, fighting, hugging, laughing, listening_to_music, running, sitting, sleeping, texting, using_laptop. KTH classes: boxing, handclapping, handwaving, jogging, running, walking. Flat KTH filenames are auto-organized into per-class folders.

3. Methodology

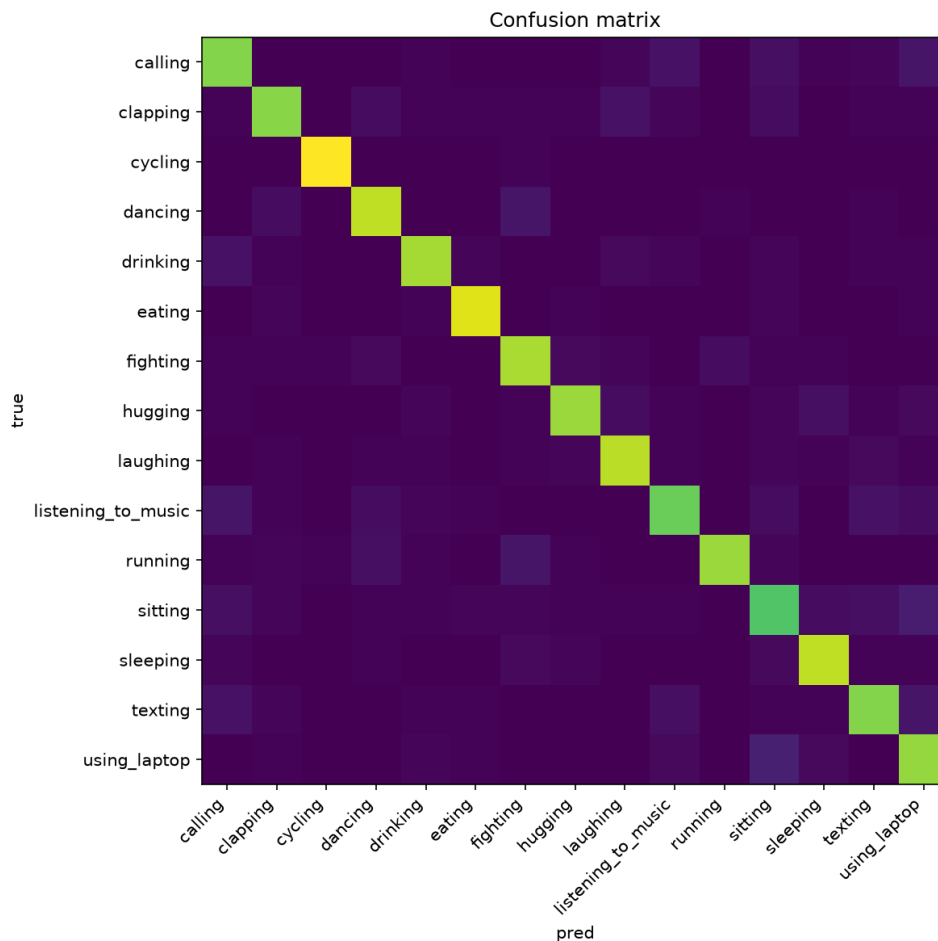
- 2D: ImageNet-pretrained EfficientNet-B3 @300px, head replaced; AdamW, cosine schedule, label smoothing 0.1, AMP mixed precision, RandomErasing/RandAugment/MixUp options; resolution stored in checkpoint and honored by all inference paths.
- 3D: R3D-18 on 16-frame 112px clips; sliding-window inference with probability averaging. (Torchvision S3D was tested and removed: its final pooling crashes on 112px inputs.)
- Detection: YOLOv8n person boxes -> 2D crop classification -> annotated frame; full-frame fallback; tunable confidence.
- App: dark Gradio UI with verdict banners, per-person table, examples, lite mode.

4. Results

Model	Epochs	Accuracy	Note
MobileNetV3-Small	12	74.02%	baseline
EfficientNet-B0	25	82.75%	mid-step
EfficientNet-B3 @300	23	85.24%	champion
R3D-18 (KTH val)	10	91.6%	video branch

Strongest 2D classes (F1): cycling 0.984, eating 0.949, dancing 0.887. Hardest: sitting 0.706, calling 0.777, texting 0.789 - visually near-identical poses plus label noise, a data ceiling rather than a pure modeling gap. 95% top-1 is therefore not a realistic target on HAR-15; the honest path (B3 @300px + aug + TTA + ensemble) points to the high-80s/low-90s.

Confusion matrix - EfficientNet-B3 (test set)



5. Reproduce

pip install -r requirements.txt + CUDA torch (cu128) + numpy<2. Download HAR-15 and KTH via src.download_data, train with src.train / src.train_video, evaluate with src.evaluate (--tta), launch with python -m src.app. Full commands are in docs/PROJECT_DOCUMENTATION.md.

- python -m src.infer --image photo.jpg | --video in.mp4 --out out.mp4 | --webcam
- python -m src.yolo_action --source 0 --out out.mp4
- python -m src.inference_video_3d clip.avi -o out_3d.mp4
- Deploy: Hugging Face Spaces (free), --share link, or Docker on a GPU VM.

6. Limitations and Future Work

- 2D sees single frames; 3D covers only 6 KTH actions.
- Small/occluded YOLO crops inherit 2D confusions.
- Next: B3 @300px + ensemble, cross-subject KTH eval, pose branch, ONNX export.